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ORIGINAL ARTICLE

Effect of newly required American football headgear on thermoregulation during exercise in the heat

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Abstract
The Guardian Cap (GC) football helmet cover is designed to reduce head impact severity, but the influence on body temperature during exercise heat stress is unknown. We determined whether the GC NXT accentuates hyperthermia in a thermal manikin model and exercising humans ($n = 12$ males, 18–35 yrs, > 91 kg). The manikin assessed heat loss, heat absorption, thermal resistance, and evaporative resistance in four helmets with and without the GC NXT. The human study recorded rectal (T_r) and skin (T_{sk}) temperature, sweat rate, heart rate (HR), thermal sensation (ThS), rate of perceived exertion (RPE), and fatigue during five 90 min exercise trials in the heat (two helmet models each with and without white and black GC NXT's). In the manikin, heat loss was lower by 2.35 W/m^2 with all helmets when the GC NXT was worn ($p = 0.0283$). The effect of the GC NXT was small compared to the variation between helmet models (Hedges' $g^* = 0.13$). T_r, T_{sk}, HR, and sweat rate were similar between trials in humans. Helmet temperatures were 5–10 °C lower with the GC NXT compared to the bare helmet ($p < 0.001$). ThS was lower with the white ($p = 0.007$) and black ($p = 0.011$) GC NXT, but RPE and fatigue were similar. Minor differences in heat loss and absorption were found in the manikin, but did not translate into altered thermoregulatory parameters in humans during intense exercise heat stress. The GC NXT did not significantly alter exertional hyperthermia, which can inform decision-making for product use.

Keywords Thermoregulation · Hyperthermia · American football · Textiles · Protective equipment

1 Introduction

Hyperthermia during exercise in the heat reduces work capacity and threatens physiological homeostasis [1]. The addition of clothing and other gear can impair thermoregulation, exacerbate cardiovascular and thermal strain, and decrease the body's innate cooling capacity [2]. American football requires substantial clothing and gear during training and competitions, which contributes to the relatively high rates of exertional heat illness in this population [3–5]. Wearing a full American football uniform accelerates the rise in core temperature, shortens the time to exhaustion [6], and increases evaporative resistance twofold compared to a t-shirt and shorts [7]. Intrinsic factors common to many American football players, such as high fat mass, low body surface area-to-mass ratios (in larger players such as linemen), and high rates of metabolic heat production [8, 9] also contribute substantially, and when combined with uniforms impeding heat loss increased

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risk for exertional heat illness. Thus, when considering new equipment in American football, thermoregulatory consequences should be considered alongside potential for injury prevention.

The helmet is particularly problematic, because it impedes evaporative heat loss [7]. The head plays a significant role in thermoregulation due to its dense cutaneous vasculature resulting in heat losses greater than other regions when normalized for surface area [10–12]. Sweat rates on the head show regional variability, being the highest on the forehead and lateral portions compared to the top or the rear [13]. This could be relevant for helmet design and thermoregulation in American football, since ventilation patterns differ between models. More studies are needed in this population given their propensity to incur severe hyperthermia and fluid losses [9, 14].

The Guardian Cap NXT (Guardian Cap NXT, Guardian Sports, Peachtree Corners, GA, USA) is a soft-shell helmet cover made of vinyl nitrile and EVA foam that attaches onto the outside of a football helmet. It is one of several aftermarket helmet add-on products designed to reduce impact severity by providing additional padding. In 2020, the Guardian Cap NXT (GC NXT) was permitted for use in National Football League (NFL) practices after laboratory testing showed an average of 9% reduction in head acceleration response metric (HARM) over bare helmets [15]. In 2022, the NFL required the use of the GC NXT for linemen, tight ends, and linebackers during preseason practices until the second preseason game, then extended that requirement to include running backs and for practices throughout the 2023 preseason and for contact practices in the 2023 season thereafter. It should be noted that the GC NXT is a product designed specifically for NFL players that is larger (369 g vs. 200 g) than the standard GC XT that is commonly used at lower levels of play. Given the novelty of the GC NXT, its potential effects on thermal exchange and core temperature during exercise have not previously been thoroughly studied. Furthermore, it was not known how the presence of the GC NXT might interact with the thermoregulatory design of the helmet, for example by blocking vent holes in the helmet or by changing its exterior color. In this study, we investigated whether the GC NXT influences thermoregulation during exercise heat stress using both thermal manikin and human volunteer testing.

2 Methods

The potential of the GC NXT helmet add-on to contribute to heat-related illness in American football players was evaluated in two ways. The first round of testing measured evaporative resistance, insulation, and the predicted heat loss of various helmet/cap combinations using the sweating surrogate head at. The second round of testing was performed at on live human volunteers to evaluate the effect of the GC NXT during physical activity. The surrogate and volunteer tests both evaluated black and white versions of the GC NXT installed on a white Riddell SpeedFlex (R41195) football helmet. The surrogate test also included black and white versions of the Schutt F7 UR2 (308300), the Schutt Air XP Pro Q11 LTD (788600), and the VICIS ZERO2-R (ZERO2-R) helmets. Using both passive surrogates and active volunteers in two different test series provided complementary evaluations of the GC NXT. Because the same equipment was used in both rounds of testing, it was possible to determine the influence of the GC NXT in thermoregulation relative to other factors, such as helmet design and color.

2.1 Manikin testing

A sweating head test was performed using the sweating head manikin system at the Textile Protection and Comfort Center, North Carolina State University, which simulates heat and sweat production by pumping preheated water to mock sweat glands distributed across the surface of the head. The head is divided into nine sections, each with its own sweating, heating, and temperature measurement systems. A temperature control unit reads the mean temperature from a Resistance Temperature Detector embedded in each section. The surrogate can be configured to achieve continuous temperature control by adjusting local heaters for each head segment to achieve a target temperature set point. The sweating surrogate head was used in two types of tests: heat loss and radiant heat absorption.

Heat loss. Eight headwear systems (22–24′ circumference, ~ 1.8 kg) were submitted for the evaluation of heat loss: four bare helmets (the Schutt Air XP Pro Q11 LTD (788600), the Riddell SpeedFlex (R41195) the Schutt F7 UR2 (308300), and the VICIS ZERO2-R helmets), and a duplicate of each helmet with a GC NXT attached. The selected helmet models utilized different padding materials and shell designs with different vent holes, allowing

Table 1 Testing conditions for the measurement of thermal and evaporative resistance of headwear systems on the sweating head manikin system

	Thermal resistance	Evaporative resistance
Air temperature (°C)	20	35
Relative humidity (%)	50	40
Air speed (m/s)	0.4	0.4
Skin temperature (°C)	35	35

observation of how helmet designs impact heat retention. The Riddell SpeedFlex and the Schutt F7 helmet platforms are currently the most commonly used by NFL players. The SpeedFlex has several vents that are open to the exterior on almost all models. The F7 models have minimal vent holes, and most are partially covered by interior padding.

Thermal and evaporative resistance of each headwear system was measured using practices adapted from the American Society of Testing and Materials (ASTM) F3426 Standard Test Method for Measuring the Thermal Insulation of Clothing Items Using Heated Manikin Body Forms and the ASTM F2370 Standard Test Method for Measuring the Evaporative Resistance of Clothing Using a Sweating Manikin, respectively [16, 17]. Testing for thermal and evaporative resistance was performed in room-temperature conditions and in high heat and moderate humidity (Table 1). Heat transfer is a measure of heat flow from the head manikin surface through the sample into the test environment. Heat loss parameters calculated from thermal transport measurements include: total thermal resistance (R_t), total evaporative resistance (R_{et}), intrinsic thermal resistance (R_{cl}), intrinsic evaporative resistance (R_{ecl}), total insulation value (I_t), permeability index (i_m), and calculated heat loss potential ($Q_{calculated,T,RH}$). Equations for the calculation of these parameters are included in the supplemental material. Calculated heat loss potential ($Q_{calculated,T,RH}$) was the metric used to evaluate the helmet and GC NXT.

Radiant heat. The sweating head manikin system was also used to observe the effect of radiant heat by measuring the temperature rise under the helmet while exposed to a solar lamp. Four headwear configurations were tested: a black Riddell SpeedFlex, a white Riddell SpeedFlex, a black GC NXT on top of a white SpeedFlex, and a white GC NXT on top of a white SpeedFlex. The thermal head was placed under a solar lamp in a controlled climatic chamber (Table 2). In these tests, the surrogate was set to have no sweating or active temperature control. The initial skin temperature of the headform was 33 °C and the temperature of each head section was measured over time, every minute for 60 min, for each headwear configuration. Maximum forehead, crown, and weighted mean skin temperatures were recorded via resistance temperature detectors (RTD's) embedded in the head. Analog signal inputs from separate RTD's are integrated by a process control unit for continuous temperature control. The RTDs are used instead of point sensors, because they provide equally weighted temperature measurements from each area. Each RTD is embedded just below the surface and provides an average temperature for each section.

Table 2 Testing conditions for the measurement of radiant heat of headwear systems on the sweating head manikin system

	Thermal resistance
Air temperature (°C)	32
Relative humidity (%)	40
Air speed (m/s)	0.6
Skin temperature (°C)	33
Solar intensity (W/m^2)	850
Sweating rate	NA

Statistical analysis: Manikin study. Two-way analysis of variance (ANOVA) was used to evaluate the effect of helmet model and GC NXT presence on heat absorption in the first study. One-way ANOVA was used to separately evaluate the effects of GC NXT presence and color on radiant heating in the second study. Two-tailed paired t tests were also performed on the same data sets. The a-priori significance value was set at 0.05. Effect size was evaluated using Hedges' g^* (0.25 = small, 0.5 = medium, 0.8 = large).

2.2 Human study

A separate study was conducted to evaluate the effect of the GC NXT on the heat loss and radiant heat balance in human subjects. All study protocols and procedures were reviewed, evaluated, and given Institutional Review Board approval at the University of Connecticut.

Participants. Twelve males between the ages of 18–35 years were recruited to participate, with body mass more than 91 kg to best represent the sample population of interest [18]. Exclusion criteria included: current musculoskeletal injury; chronic health condition affecting thermoregulation; reported use of any supplements or medications affecting thermoregulation (e.g., anticholinergics and amphetamines); and a history of cardiovascular, metabolic, and/or renal disease. All participants were required to provide written informed consent prior to beginning any study-related procedures.

Study design. A randomized cross-over design was used where each participant completed a total of five exercise trials in an environmental chamber (CANTROL, Environmental Systems Limited, Ontario, Canada) at 40 ℃, 40% relative humidity, ~ 800W/m^2 solar radiation (Super PAR CP60 EXC 240 V 1000 W lamps). Research staff utilized a potentiometer to regulate the intensity of solar radiation and ensure that the solar radiant level was set at 800 W/m^2. Four combinations of football helmet type (Riddell SpeedFlex and Schutt F7 UR2) and GC NXT color (white and black) were tested, and a control trial was done with the Riddell SpeedFlex helmet only, which was white

in color. Experimental trials were not restricted to a specific time of day, but each participant completed their subsequent four trials at the same time of day (within an hour) as their first trial and rested a minimum of 48 h between visits. Exercise was completed on a treadmill (COSMED T150DE, Rome, Italy) set at 1% grade with 15-min bouts of exercise, each consisting of a 5-min walk at 30–40% velocity at $\dot{V}O_{2\,max}$ ($v\dot{V}O_{2max}$), a 5-min jog at 50–60% $\dot{V}O_{2max}$, and five sets of a 20-s run with 40 s recovery at 80–100% $\dot{V}O_{2max}$. These 15-min workouts were repeated continuously until 90 min was reached, volitional exhaustion occurred, or rectal temperature (T_r) reached 40 °C. Ninety minutes was established as the upper time limit, since most American football players reach peak core temperatures during practice in ≤ 90 min (14) and a T_r of 40 °C was predetermined as the ethical upper limit for the study according to standard laboratory practices.

Measurements and outcomes. Participants' anthropometrics, body composition (COSMED BOD POD, Rome, Italy), and aerobic capacity were measured on the baseline visit. Body surface area (BSA) was calculated from mass and height [19]. Following a 10-min warm-up, participants completed a treadmill test for maximal oxygen consumption ($\dot{V}O_{2max}$) and heart rate (HR_{max}) where the grade was constant at 1%, velocity increased in ~ 0.8 km/h increments every two minutes until volitional exhaustion, and gas exchange measurements were collected continuously (TrueOne 2400; Parvo Medics, Salt Lake City, UT, USA). System calibration prior to each session was performed per the manufacturer's instructions using a standard room air gas mixture of O_2 and CO_2 and a 3L syringe over various flow rates. $\dot{V}O_{2max}$ was confirmed by a plateau in O_2 consumption despite increasing workload. Heart rate was monitored continuously using telemetry (H10, Polar Electro Oy, Kempele, Finland). If a participant did not complete the final 2-min stage, the peak workload (WR_{max} or $v\dot{V}O_{2max}$) was prorated [20].

Prior to experimental trials, euhydration was confirmed with urine specific gravity (USG ≤ 1.025) and nude body mass was measured (Defender® 7000XtremeW; OHASUS Corp., Parsippany, NJ, USA). Participants donned a heart rate monitor and self-inserted a rectal thermistor (YSI, Zoll Medical Corporation, Chelmsford, MA, USA) 10–12 cm beyond the anal sphincter. Rectal temperature (T_r) was monitored continuously (MP160, BIOPAC Systems Inc., Goleta, CA, USA). Four calibrated surface thermistors (iButtons, 1921H, Lawrenceburg, KY) for skin temperature (T_{skin}) were secured in place with tape (Transpore, 3 M, London, ON, Canada) to the chest, lateral deltoid, anterior thigh, and posterior lower leg on the right side of the body and Ramanathan's four site method was used to calculate mean T_{skin} [21]. Two additional thermistors were placed on the face at

Table 3 Average heat parameters for all head manikin zones

Sample	R_t (°C*m²/W)	R_{et} (kPa*m²/W)	R_{cl} (°C*m²/W)	R_{ecl} (kPa*m²/W)	I_t (clo)	i_m	$Q_{calculated}$ 25 °C, 65% RH (W/m²)
Q11	0.1261	0.0233	0.0603	0.0125	0.81	0.33	232.3
SpeedFlex	0.1411	0.0276	0.0752	0.0169	0.91	0.31	200.1
UR1	0.1237	0.0211	0.0576	0.0104	0.80	0.36	249.5
ZERO2	0.1653	0.0313	0.0990	0.0205	1.07	0.32	174.4
Q11 + GC	0.1315	0.0238	0.0653	0.0131	0.85	0.34	225.6
SpeedFlex + GC	0.1489	0.0282	0.0831	0.0174	0.96	0.32	193.5
UR1 + GC	0.1245	0.0217	0.0587	0.0109	0.80	0.35	244.9
ZERO2 + GC	0.1625	0.0321	0.0963	0.0213	1.05	0.31	172.9

the cheek (T_{face}) and on the interior surface of the helmet (T_{helmet}).

Solar radiation was simulated during all trials with portable lamps (Super PAR CP60 EXC-240V 1000W) that contained a potentiometer to regulate radiation levels. The radiation level was measured at the start of each trial using a solar power radiation meter placed on the crown of the helmet and set at ~ 800W/m^2, a typical level under clear sky conditions. The lamps were placed at ceiling level and directed downward at the helmet/upper back area of the participant. Participants were provided sunscreen (SPF-100) as a safety precaution to any exposed areas of skin. A total of 10 mL/kg body weight of fluids were provided throughout the trial, split into three separate boluses with the first given after the first 15 min. This volume was chosen based on previously reported sweat rates in American football players (> 2L/h) [9]. With an expected mean body mass of ~ 100 kg, this would replace ~ 50% of fluid losses which is typical in field studies of this population [22]. Participants were encouraged to consume the total volume prior to completion of the trial. A urine sample was collected after trial completion, and nude body mass measured to determine body mass loss and sweat rate. Subjective measures of thermal sensation (ThS) [23], rating of perceived exertion (RPE) [24], and fatigue [25] were collected every 15 min.

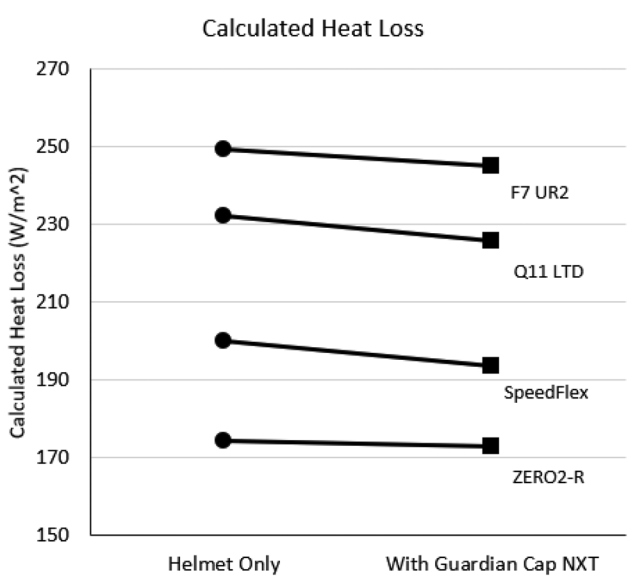

Fig. 1 Calculated heat loss by helmet model in the manikin study with and without the GC NXT, plotted to show differences in helmet model

Statistical analysis: human study. Subject characteristics were analyzed with descriptive statistics (mean ± SD). One-way repeated-measures ANOVA was used to detect differences between conditions for trial duration, end-exercise T_r, T_r rate of rise, and HR (average and peak). Two-way repeated-measures ANOVA (condition x time) was used to detect differences in T_r, mean T_{sk}, T_{helmet}, and T_{face} at 15-min intervals over the trial duration. If sphericity was violated, a Greenhouse–Geisser correction was used. If an interaction was observed, pairwise comparisons detected where the differences occurred, and Bonferroni adjustment corrected for multiple comparisons. Subjective data (ThS, fatigue, RPE) were analyzed at 45 and 75 min using Friedman's test and Wilcoxon Signed-Rank for post hoc analysis when necessary.

3 Results

3.1 Manikin testing

The average heat parameters for all manikin zones calculated from the heat loss study were tabulated (Table 3). Calculated heat loss ($Q_{calculated}$) was lower with the GC NXT for all helmets ($p = 0.0283$). On average, the GC NXT resulted in a 4.85 W/m^2 reduction (2.3%) in heat loss. Although adding a GC NXT to a standard football helmet significantly reduced the amount of heat loss from the head, the effect was small compared to the variation in helmet design (Hedges' $g^* = 0.13$). For example, the F7 UR2 with a GC NXT installed had less heat loss than the F7 UR2 without a GC NXT (244.9 W/m^2 vs. 249.5 W/m^2); however, it was still greater than the SpeedFlex with (193.5 W/m^2) or without (200.1 W/m^2) a GC NXT (Fig. 1).

In the radiant heat study, the black helmet without the GC NXT resulted in the highest maximum temperatures at the forehead (35.5 °C) and crown (36.0 °C), as well as mean head temperature (34.3 °C) (Table 4). Using the bare white helmet as a baseline, the bare black helmet increased (+ 0.59 °C) the weighted mean temperature, while the black

Table 4 Max forehead, crown, and mean head average temperature for the human study

Sample	Average max forehead (°C)	Average max crown (°C)	Max weighted mean head (°C)
Black SpeedFlex	35.46	35.97	34.28
White SpeedFlex	34.32	34.38	33.69
White SpeedFlex + Black GC	34.23	34.35	33.67
Black SpeedFlex + White GC	33.86	33.93	33.48

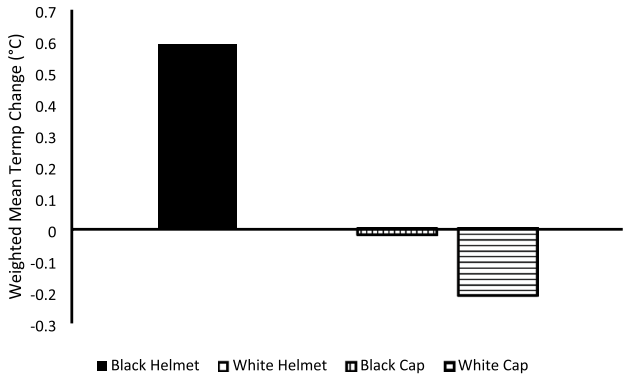

Fig. 2 Maximum recorded head temperature changes, recorded as weighted means across all nine surface measurement sections, during the radiant heat manikin study relative to the baseline white helmet

and white GC NXT installed on the white helmet decreased (− 0.02 °C and − 0.21 °C) temperature (Fig. 2). The GC NXT decreased radiant heat absorption when compared to the bare helmet ($p < 0.02$). For both the GC NXT and bare helmet comparisons, the lighter color also resulted in a decrease in radiant heat absorption ($p < 0.04$). The effects of both exterior color and GC NXT presence were large (Hedges' $g^* = 1.3 – 1.6$) and similar in magnitude. Only one helmet design (Riddell SpeedFlex) was used in the radiant heat study, so the effects of color and GC NXT presence could not be compared across different helmet designs.

3.2 Human study

Twelve participants were recruited and enrolled between January and March 2023 and their characteristics can be found in Table 5. Solar radiation levels averaged 830 ± 11 W/m^2 measured at the helmet and 366 ± 12 W/m^2 at the base of the neck. There was no difference in trial duration between headgear conditions (range: 80–86 min, $p = 0.393$). The number of participants reaching 90 min ranged from six to nine (RC: 9/12; RB: 7/12; RW: 7/12; FW: 7/12; FB: 6/12). Early termination was due to a combination of pre-syncopal symptoms and reaching a T_r of 40 ℃. Because so many participants did not complete the full 90 min, outcome variables were analyzed over time (T_r, mean T_{sk}, T_{helmet}, and T_{face}) only up to 75 min.

Core, skin, and helmet temperatures. Rectal temperature (T_r) increased over time in all trials ($p < 0.001$). There was no effect of headwear configuration ($p = 0.731$) or a headwear configuration × time interaction ($p = 0.302$) (Fig. 3A). End-exercise rectal temperature (T_r) (Fig. 3B) ($p = 0.401$), change in temperature (ΔT_r) (Fig. 3C) ($p = 0.445$), and rate of temperature (T_r) rise (Fig. 3D) ($p = 0.861$) were also not different between headwear configurations. Mean skin (T_{skin}) and face (T_{face}) temperatures (Fig. 4A, B) increased abruptly in all conditions and rose slightly throughout the remainder of the trial (main effects for time: $p < 0.001$), but there were no effects of headwear configuration (mean T_{sk}: $p = 0.0215$; T_{face}: $p = 0.315$) or of headwear configuration × time interactions (mean T_{sk}: $p = 0.572$; T_{face}: $p = 0.810$). For helmet temperature (T_{helmet}) (Fig. 4C), there was a main effect of time ($p < 0.001$), headwear configuration ($p < 0.001$), and a headwear configuration × time interaction ($p < 0.001$). The control condition (RC) had higher helmet temperatures (T_{helmet}) than all other conditions up to 30 min (all $p < 0.022$), was significantly higher than RB ($P = 0.006$), RW ($P = 0.017$), and FW ($p = 0.029$) at 45 min, and significantly higher than RB at 60 ($P = 0.005$) and 75 min ($p = 0.044$).

Heart rate. There was no effect of helmet condition on average heart rate (RC: 151 ± 12; RB: 149 ± 13; RW: 146 ± 16, FW: 150 ± 10, FB: 147 ± 16 bpm) ($p = 0.673$), peak heart rate (RC: 185 ± 12; RB: 182 ± 11; RW: 180 ± 18, FW: 187 ± 12, FB: 182 ± 16 bpm) ($p = 0.359$), or percentage of maximum heart rate (RC: 96 ± 3; RB: 95 ± 4; RW: 95 ± 6, FW: 97 ± 4, FB: 94 ± 5% HR_{max}) ($p = 0.367$) achieved during the trial.

Fluid consumption, body mass loss, and sweat rate. Fluid consumption (RC: 0.93 ± 0.26L; RB: 0.80 ± 0.33; RW: 0.84 ± 0.19, FW: 0.91 ± 0.28, FB: 0.83 ± 0.26 L) (p = 0.224), percent body mass loss (RC: 1.24 ± 0.57; RB: 1.24 ± 0.43; RW: 1.29 ± 0.47, FW: 1.45 ± 0.54, FB: 1.34 ± 0.59%) (p = 0.441), and sweat rate (RC: 1.82 ± 0.34; RB: 1.79 ± 0.5; RW: 1.88 ± 0.42, FW: 2.03 ± 0.48, FB: 1.91 ± 0.51 L/hr) ($p = 0.180$) were not statistically different between conditions.

Table 5 Subject characteristics (mean ± SD). FFM: fat free mass; FM: fat mass; $\dot{V}O_{2max}$ (mL/kg/min): relative maximal oxygen consumption; $\dot{V}O_{2max}$ (L/min): absolute maximal oxygen consumption; RHR: resting heart rate; HR_{max}: maximal heart rate; BSA: body surface area

	Age	Height (cm)	Weight (kg)	FFM (kg)	FM (kg)	Body Fat (%)	$\dot{V}O_{2max}$ (mL/kg/min)	$\dot{V}O_{2max}$ (L/min)	RHR (bpm)	HR_{max} (bpm)	BSA (m^2)
Mean	26	185.7	102.2	78.9	23.1	22.2	41.8	4.2	76	192	2.25
SD	6	6.2	11.9	9.1	9.9	8.1	4.9	0.5	10	10	0.14

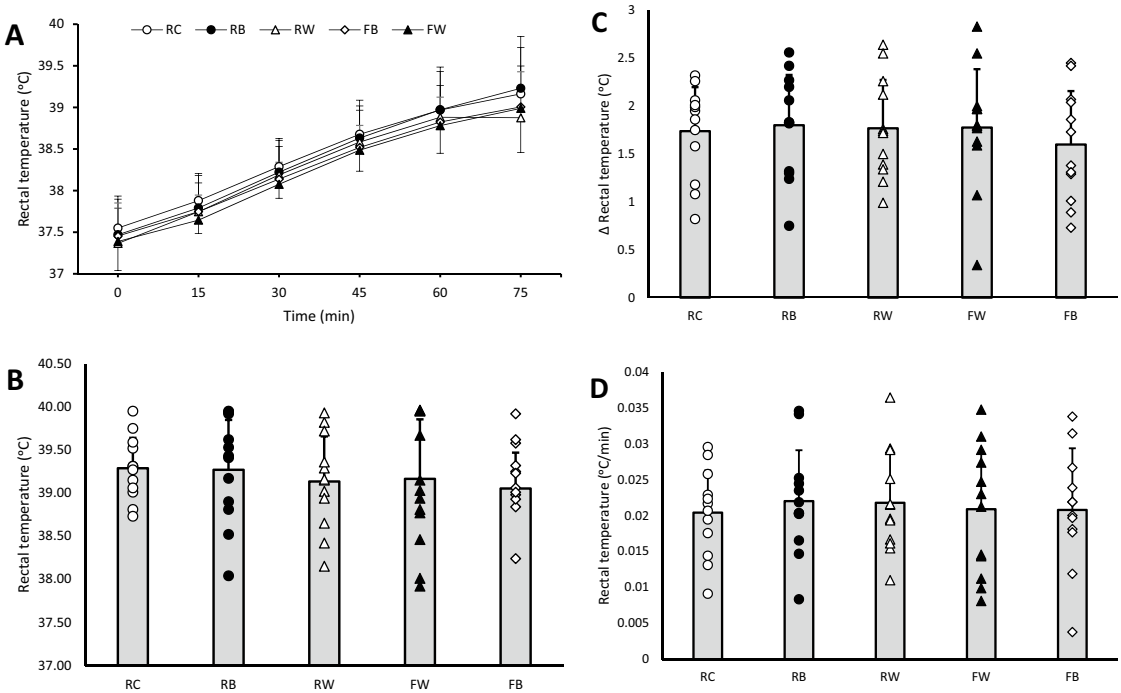

Fig. 3 **A** Rectal (T_r) temperature over time. **B** End-exercise T_r. **C** Absolute change in T_r from the start to end of the trial. **D** Rate of rise in T_r (°C/min). RC: Riddell SpeedFlex (control, no GC NXT); RB: Riddell SpeedFlex + black GC NXT; RW: Riddell SpeedFlex + white GC NXT; FW: Schutt F7 + white GC NXT; FB: Schutt F7 + black GC NXT

Subjective measures. The rating of perceived exertion (RPE) ($p = 0.149$), subjective measures of thermal sensation (ThS) ($p = 0.919$), and fatigue ($p = 0.510$) were similar between headwear configurations at 45 min. Differences in the rating of perceived exertion (RPE) were detected at 75 min ($p = 0.038$), but post hoc analysis failed to reveal any pairwise differences (all $p > 0.102$). Thermal sensation at 75 min was slightly higher in RC versus RW ($p = 0.007$) and RB versus RW ($p = 0.011$). Differences in fatigue were detected at 75 min ($p = 0.022$), but subsequent post hoc testing found no pairwise differences (all $p > 0.131$).

4 Discussion

This research sought to comprehensively evaluate the thermal effects of the GC NXT football helmet cover by conducting complementary studies using test manikins and live human subjects. The manikin studies found that the GC NXT created two competing effects. On the one hand, the GC NXT modestly reduced heat loss compared to each helmet model alone through an insulation effect that would apply to any clothing. The effect size of the GC NXT on heat loss was small compared to the effect of helmet model (Hedges' $g^* = 0.27$). An opposite effect was observed in the radiant heat study. The GC NXT reduced radiant heat absorption, potentially through reflection of thermal radiation, its material properties (specific heat, thermal conductivity), or alterations to convective heat flow and insulation. This reduction in radiant heat absorption from adding the GC NXT was not statistically different than changing the exterior color of the helmet/cap from black to white. The manikin test results suggested that the GC NXT could potentially have small positive or negative thermal effects on an athlete depending on the weather conditions.

The human volunteer study, on the other hand, found no significant objective thermoregulatory or cardiovascular differences when exercising between any of the various headwear configurations. Temperature (T_r, mean T_{sk}, T_{face}), average heart rate, peak heart rate, $\%HR_{max}$, sweat rate, and body mass loss were unchanged between headwear configurations. This finding is important, because the headwear configurations varied not only in whether or not a GC NXT was used, but also in exterior color and helmet design. In the manikin tests, exterior color had an effect size comparable to the GC NXT and helmet design had an effect size several times greater than the GC NXT. The fact that none of these parameters were significant in the human study, even in combination, indicates that the changes in head and helmet temperature caused by differences in equipment are too small to have significant systemic effects on athlete, even

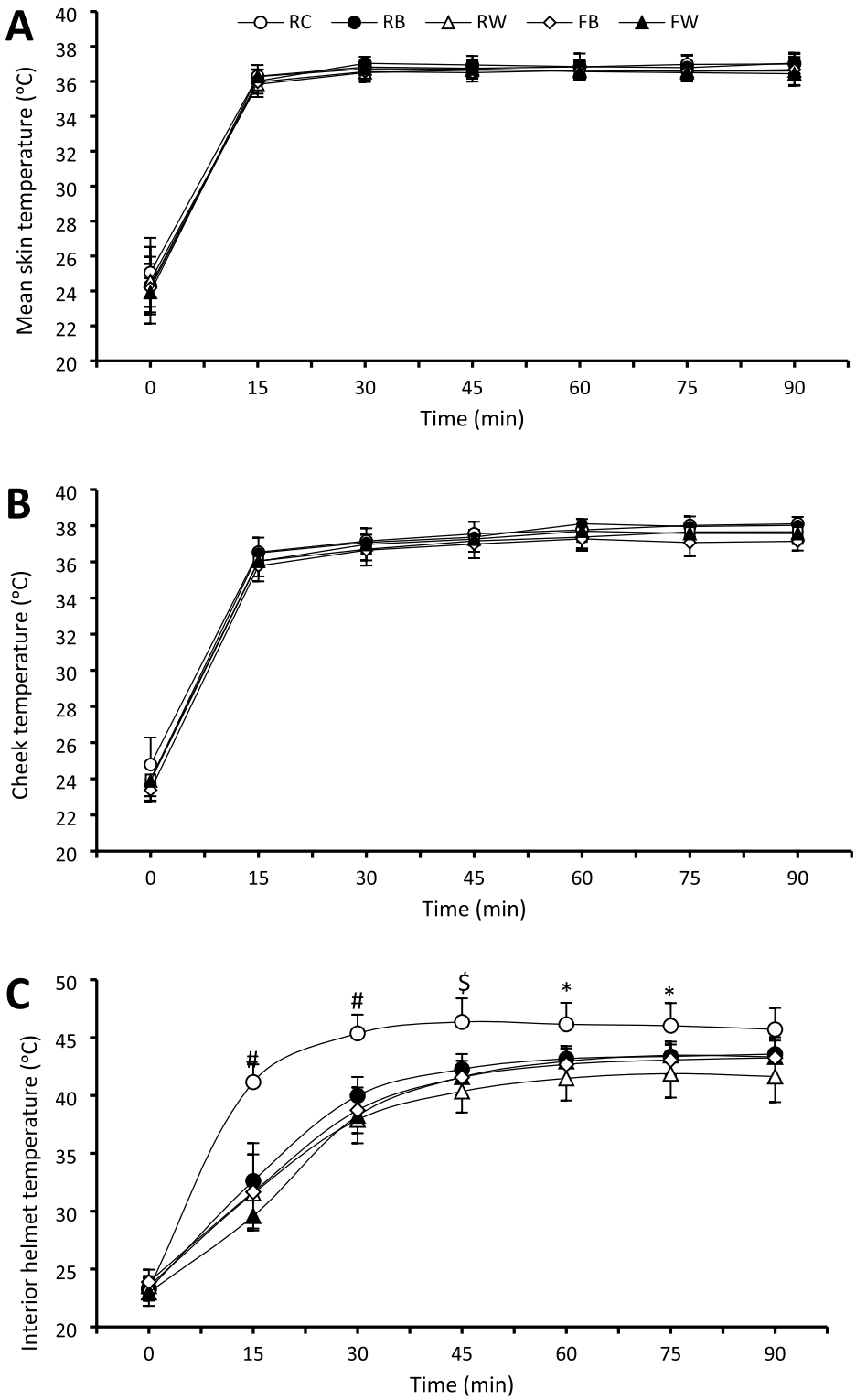

Fig. 4 **A** Mean skin temperature over time. **B** face temperature over time. **C** interior helmet temperature over time. # RC > all; $ RC > RB, RW, FW; * RC > RB

during intense exercise just below the typical thresholds for symptoms of heat illness.

Although objective physiological measurements did not uncover any differences between headwear configurations in the human study, the volunteers did report being hotter toward the end of the trial (75 min) in the control condition and with the Riddell SpeedFlex covered by a black GC NXT compared to the Riddell Speedflex covered by a white GC NXT. The manikin study, performed in a simpler and less variable environment where white GC NXTs were associated

with statistically significant lower radiant heat absorption, supports this trend, though available physiological data do not indicate a specific explanation in the more variable environment of the human subject study; however, the authors cannot discount the possibility of a placebo effect, since participants were not blinded to the caps' color. The interior helmet temperature (T_{helmet}) on the interior crown of the helmet was ~ 5–10 °C lower when the GC NXT was worn (Fig. 4), possibly due to the caps reflecting some of the radiant heat as they did in the manikin tests. The radiant heat lamps in the human tests were positioned to replicate outdoor conditions midday (i.e., overhead sunlight targeting the head/neck). Differences in helmet temperature due to the GC NXT were most pronounced in the first 30 min of exercise and decreased as exercise progressed, probably due to contributions of heat radiating from the head and warming the inside of the helmet over time. For practical reasons, forehead or crown temperature was not measured because of discomfort/pain associated with skin thermistors being placed between the head and helmet. Face temperature was measured on the cheek instead, but the differences in T_{helmet} between conditions did not influence T_{face} which was identical between all conditions (Fig. 4).

5 Conclusions

To summarize the combined findings of the manikin and volunteer studies, there is no objective advantage or disadvantage pertaining to thermoregulation when wearing the GC NXT during exercise in the heat under laboratory-simulated conditions. The thermal effects of the GC NXT are small and may be positive or negative depending on environmental conditions. The GC NXT can reduce the temperature inside the helmet and the subjective sensation of heat by either reflecting direct radiant heat from the sun or due to its material properties. However, the effect of the cap was small, similar to the effect of helmet color (white vs. black), and much less than the effect of helmet design. None of the differences in helmet temperature caused by variations in GC NXT use, exterior color, or helmet design in the manikin study translated into measurable changes in objective thermoregulatory or cardiovascular parameters in humans subjected to intense exercise in the heat. These findings indicate that the GC NXT does not contribute to an exaggerated thermoregulatory response during intense exercise heat stress above that of the helmet alone.

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Data availability Data from this study can be made available upon reasonable request.

Declarations

Conflict of interest Ann Good, Loghan Thomas, James Funk, and Jeff Crandall are employed and affiliated with Biocore LLC, who had a role in developing the product that was tested (GC NXT). Biocore LLC supplied the helmets and equipment that were used for testing and contracted the separate laboratories to perform the thermal manikin and human studies.

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